Grinding machine

Ambrose Webster was an American scientist; citizen of Massachusetts invented the small grinding machine in 1860.

Grinding wheel

The scientist named Suyin Palson was invented grinding wheel with mixture of emery powder, clay and water in 1873.

4.1 Introduction

Grinding is a metal cutting operation where metal is cut by a rotating abrasive wheel. This machine on which grinding the operation is performed is called a grinding machine.

Grinding is done to obtain very high dimensional accuracy and good surface finish on the work place. The accuracy of grinding process is 0.000025 mm. The amount of material removed from this process is very less.

4.2 Types of grinding machines

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According to the accuracy of the work to be done on a grinding machine, they are classified as

1. Rough grinding machines or Non-Precision grinding machines
2. Precision grinding machines

4.3 Non-Precision grinding machine

The non-precision grinding machines are used to remove unwanted projections from casting and welding, when the surface finish is not important. The main types of non-precision grinders are

1. Hand grinding machine
2. Bench grinding machine
3. Floor stand grinding machine
4. Flexible shaft grinding machine
5. Swing frame grinding machine
6. Abrasive belt grinding machine

Hand grinding machine



It is a small grinder which can be carried from one place to another place easily. A small grinding wheel is mounted at the end of motor shaft. The motor is connected by along wire to the main switch box. It is used for grinding large forgings, casting, welded joints and sheet metal work etc.

Bench grinding machine

This grinder is mounted on a bench. Two grinding wheels are connected to both ends of the motor shaft. The wheels are covered by guards. Work rest is provided for supporting the work piece while grinding.

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